

Budget Sets, Preferences & Choice : Problem Set Solutions

Problem 1

Let x_1 = pages in ECON
 x_2 = pages in SOC

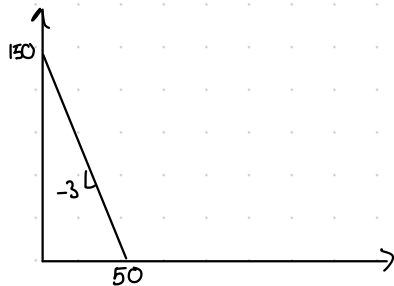
Two points: $(40, 30)$, $(30, 60) \rightarrow$ slope $m_1 = \frac{60-30}{30-40} = -\frac{30}{10} = -3$

Then the budget line is: $x_2 = 60 - 3(x_1 - 30) = 150 - 3x_1$

(a) 150 pages of SOC

(b) 50 pages of ECON

(c)



(d) For $x_1 \leq 25$, we are given two points $(10, 140)$, $(20, 130)$
 $\rightarrow$ slope $m_2 = \frac{130-140}{20-10} = -\frac{10}{10} = -1$

The first part of the budget line is:

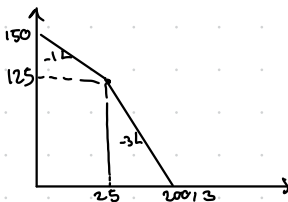
$$x_2 = 140 - (x_1 - 10) = 150 - x_1 \quad \text{for } x_1 \leq 25$$

There will be a kink at $x_1 = 25$, $x_2 = 150 - 25 = 125$

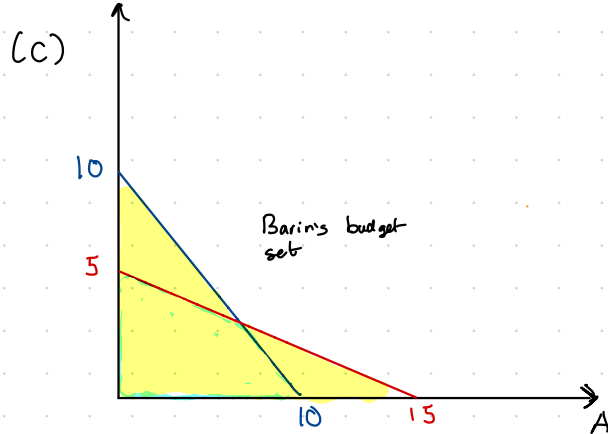
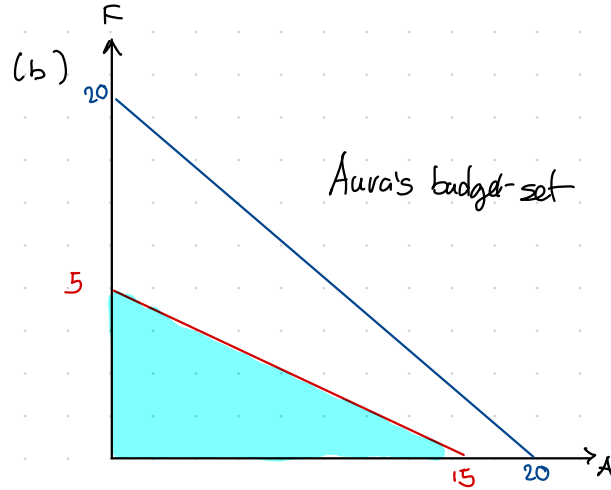
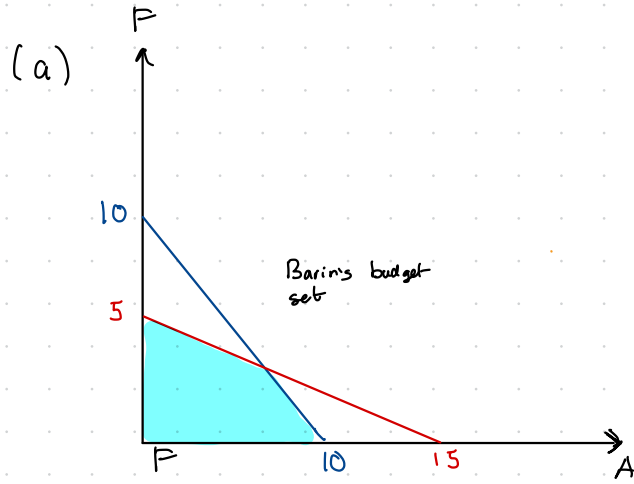
To obtain the second part we do:

$$x_2 = 125 - 3(x_1 - 25) = 200 - 3x_1 \quad \text{for } x_1 > 25$$

The x_1 intercept here is $x_1 = 200/3$



Problem 2:



(d) No, examples:

- Time & money
- Money & Calorie budget
- Money & Alcohol budget

Problem 3:

Ordering by Weight (w) $W \succ_w H \succ_w J$

Ordering by Speed (s) $H \succ_s J \succ_s W$

Ordering by Obedience (o) $J \succ_o W \succ_o H$

(a) The coach prefers W to H , i.e. $W \succ H$

(b) The coach prefers H to J , i.e. $H \succ J$

(c) The coach prefers J to W , i.e., $J \succ W$

(d) No: $W \succ H$, $H \succ J$ but it is not the case that $W \succ J$

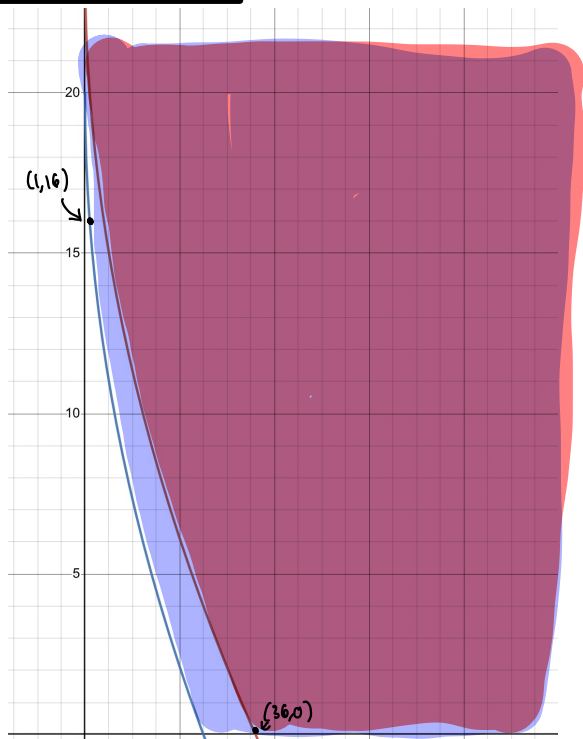
(f) No, the 3 of them are not comparable

(d) Yes, they would be transitive.

If the coach: $x \succ y$
 $y \succ z$ This means:

$$\left. \begin{array}{l} x \succ_w y \\ x \succ_s y \\ x \succ_o y \\ y \succ_w z \\ y \succ_s z \\ y \succ_o z \end{array} \right\}$$
$$\Rightarrow \left. \begin{array}{l} x \succ_w z \\ x \succ_s z \\ x \succ_o z \end{array} \right\} \Rightarrow x \succ z \quad /$$

Problem 4 (a)



(b) Yes

(c) Firstly note that $(9, 8)$ and $(4, 12)$

pass through the blue indifference curve

while $(9, 12)$ and $(4, 16)$ pass through the red one:

The slope of the two curves is given by $m = -\frac{2}{\sqrt{x}}$

Thus at the points requested, the slopes are:

$$(9, 8) \quad m_1 = -\frac{2}{\sqrt{9}} = -\frac{2}{3}$$

$$(4, 12) \quad m_2 = -\frac{2}{\sqrt{4}} = -1$$

$$(9, 12) \quad m_3 = -\frac{2}{\sqrt{9}} = -\frac{2}{3}$$

$$(4, 16) \quad m_4 = -\frac{2}{\sqrt{4}} = -1$$

→ You can explain this as the slope of the tangent line at that point.

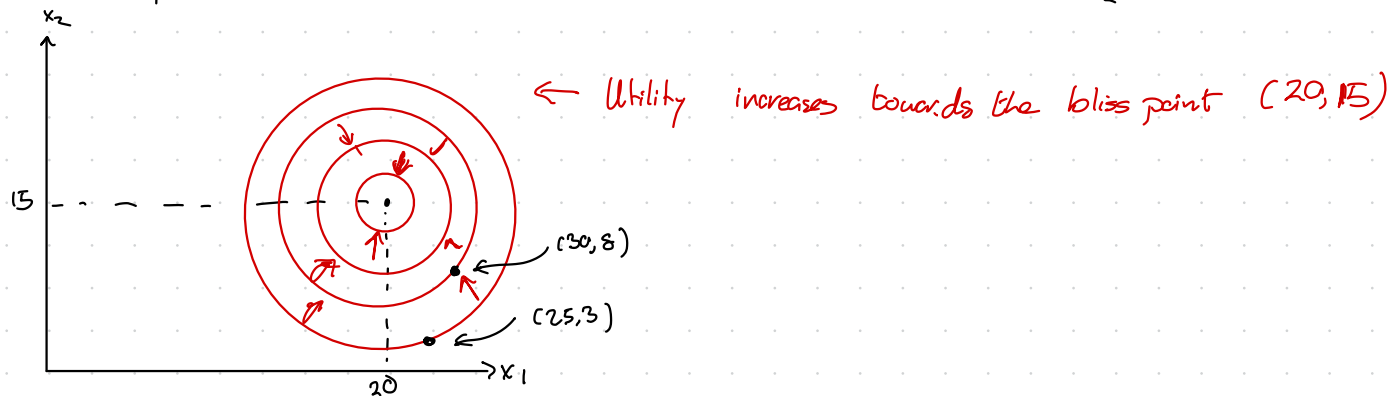
(d) Yes, they do

(e) Because of (d) yes, they represent convex preferences

Problem 5

(a) A utility function representing her preferences is $U(x_1, x_2) = -[(x_1 - 20)^2 + (x_2 - 15)^2]$

(b)



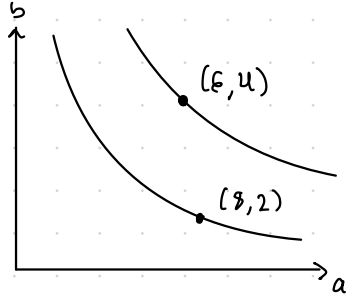
(c)
$$\left. \begin{aligned} U(25, 3) &= -169 \\ U(30, 8) &= -149 \end{aligned} \right\} \Rightarrow \text{Yes, she would prefer the combination } (30, 8)$$

$$(d) \quad MRS(25, 3) = \frac{25 - 20}{3 - 15} = -\frac{5}{12}$$

$$MRS(30, 8) = \frac{30 - 20}{8 - 15} = -\frac{10}{7}$$

Problem 6

(a) $u(8, 2) = \frac{16}{100} = 0.16$, $IC(8, 2): 0.16 = \frac{ab}{100} \Rightarrow 16 = ab \Rightarrow b = \frac{16}{a}$



$$u(6, 4) = \frac{6 \cdot 4}{100} = \frac{24}{100} = 0.24$$

$$IC(6, 4): 24 = ab \Rightarrow b = \frac{24}{a}$$

(b) $v(8, 2) = 1000 \cdot 8^2 \cdot 2^2$

$$v(6, 4) = 1000 \cdot 6^2 \cdot 4^2$$

$$IC(8, 2): 1000 \cdot 8^2 \cdot 2^2 = 1000 a^2 b^2 \\ \Rightarrow 16^2 = (ab)^2 \Rightarrow 16 = ab \Rightarrow b = \frac{16}{a}$$

$$IC(6, 4): 1000 \cdot 6^2 \cdot 4^2 = 1000 a^2 b^2 \\ \Rightarrow 24^2 = (ab)^2 \Rightarrow 24 = ab \Rightarrow b = \frac{24}{a}$$

$\therefore$ The same diagram as above.

(c) Yes for both

(d) They are the same preferences

(e) v is a monotonic transformation of u : $f(u) = 10000 u^2$